

LANVEOC POULMIC AB, France

WMO: 071090

Lat: 48.2794N

Lon: 4.4394W

Elev: 287

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.8	33.8	24.5	18.8	35.1	27.4	21.5	37.0	33.3	48.5	29.6	49.5	8.7	90	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	78.2	66.1	74.6	64.4	71.4	62.9	67.5	75.7	65.7	71.7	64.3	68.9	8.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	92.8	69.7	63.6	88.8	67.8	62.5	85.4	66.2	32.0	76.2	30.6	71.9	29.6	69.0	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.8	23.6	21.0	DB	27.7	85.1	3.1	4.4	25.5	88.3	23.7	90.8	21.9	93.3	19.7	96.4
			WB	25.8	71.2	3.4	2.6	23.4	73.1	21.4	74.6	19.6	76.1	17.1	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	45.3	45.2	47.8	50.8	55.2	60.1	63.1	62.6	60.5	55.8	50.2	46.7	(6)
	DBStd	7.86	4.87	4.74	4.62	4.36	4.35	4.37	3.90	3.56	4.24	4.43	4.36	5.15	(7)
	HDD50	621	154	142	95	39	7	1	0	0	1	8	50	124	(8)
	HDD65	4208	610	555	533	427	305	158	86	92	146	285	445	566	(9)
	CDD50	1955	9	7	27	63	168	304	405	390	316	188	55	23	(10)
	CDD65	68	0	0	0	0	2	12	26	16	11	1	0	0	(11)
	CDH74	381	0	0	0	1	15	87	156	80	38	4	0	0	(12)
	CDH80	64	0	0	0	0	1	18	26	17	2	0	0	0	(13)
(14) Wind	WSAvg	10.4	11.8	11.4	11.2	10.3	10.0	9.6	9.6	9.1	9.1	9.8	11.3	12.1	(14)
(15) Precipitation	PrecAvg	41.6	5.0	4.1	2.6	3.3	2.6	2.2	2.5	2.4	2.6	4.3	5.0	5.0	(15)
	PrecMax	52.0	9.7	9.4	7.4	7.5	5.4	5.7	5.7	5.0	6.1	7.0	9.8	9.2	(16)
	PrecMin	32.5	1.0	0.6	0.4	0.7	0.4	0.4	0.6	0.4	0.2	1.0	1.1	0.9	(17)
	PrecStd	5.4	2.4	2.1	1.6	1.9	1.4	1.4	1.3	1.4	1.6	1.5	2.2	2.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.8	66.9	71.5	75.7	82.6	82.8	82.7	78.2	69.1	61.8	57.9	(19)
		MCWB	53.5	51.2	54.0	59.2	62.9	67.9	68.4	69.1	65.5	62.7	57.9	54.6	(20)
	2%	DB	54.4	54.4	59.8	65.5	69.9	75.8	78.0	75.5	73.8	65.7	59.6	56.4	(21)
		MCWB	52.7	50.9	51.9	55.6	60.7	64.3	66.2	65.9	64.7	60.8	57.0	54.3	(22)
	5%	DB	53.3	53.0	55.9	60.9	66.0	71.6	74.3	71.0	70.1	63.7	58.0	55.2	(23)
		MCWB	51.7	50.6	50.7	53.5	58.4	62.8	64.6	63.6	62.5	59.8	55.7	53.4	(24)
	10%	DB	52.3	51.9	53.8	57.4	62.6	68.1	70.4	68.4	67.0	62.0	56.7	54.1	(25)
		MCWB	50.6	49.8	50.4	51.8	56.6	61.0	62.7	62.2	61.3	59.0	54.5	52.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	53.6	55.3	60.0	64.5	69.4	69.8	70.6	68.3	63.8	59.9	56.0	(27)
		MCDB	54.9	54.7	63.3	68.4	72.9	80.7	80.4	79.9	75.7	67.2	60.7	56.6	(28)
	2%	WB	53.4	52.5	53.5	56.5	61.4	65.7	67.3	66.8	65.7	62.2	57.7	54.9	(29)
		MCDB	54.1	53.5	56.7	63.3	68.6	73.5	76.3	72.6	70.7	64.5	58.8	55.8	(30)
	5%	WB	52.2	51.3	52.3	54.4	59.3	63.5	65.3	65.2	64.0	61.0	56.3	53.9	(31)
		MCDB	53.0	52.5	54.7	59.1	64.7	69.9	71.9	69.1	67.6	62.8	57.6	55.0	(32)
	10%	WB	51.0	50.2	51.2	52.9	57.4	61.5	63.7	63.8	62.6	59.7	54.8	52.7	(33)
		MCDB	52.0	51.7	53.3	56.4	61.2	66.4	68.5	66.8	65.4	61.4	56.4	54.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.0	10.7	12.1	12.5	12.6	12.3	12.4	12.7	10.4	8.9	8.2	(35)
		MCDBR	7.2	9.7	14.1	18.2	18.6	19.3	19.9	16.5	16.7	11.6	9.5	7.6	(36)
	5% WB	MCWBR	7.3	7.7	8.7	10.3	10.7	10.6	10.3	8.9	8.8	7.8	8.0	7.4	(37)
		MCWBR	7.0	7.8	11.0	14.8	16.0	17.2	17.3	13.8	14.0	9.9	8.8	7.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.335	0.349	0.374	0.382	0.385	0.379	0.379	0.369	0.358	0.356	0.337	0.328	(40)
		taud	2.399	2.381	2.329	2.325	2.362	2.405	2.406	2.456	2.467	2.459	2.426	2.368	(41)
	Ebn at Noon		230	254	266	275	279	280	278	277	269	249	227	214	(42)
		Edh at Noon	22	28	34	37	37	36	36	32	29	25	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		310	570	949	1485	1757	1901	1810	1553	1239	728	389	259	(44)
	RadStd		23	57	99	122	151	199	95	124	88	54	30	30	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+2.70	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page